PhilaLink: Project Specification

"Linking Communities to Healthier Lives"

A Comprehensive Digital Health Ecosystem for South African Communities

1. Executive Summary & Vision

PhilaLink is a community-centric healthcare solution designed for the South African public health landscape. It addresses the critical challenge of chronic medication adherence and diagnostic accessibility in underserved areas. The system digitizes the manual "Proxy" collection system (CCMDD), provides automated health reminders, and leverages AI for preliminary symptom assessment.

> **The "Mom" Story:** Inspired by community health caregivers who manage medication collection for multiple elderly citizens, PhilaLink replaces physical collection cards with a digital dashboard, eliminating the risk of missed dates and nurse-patient friction.

2. Technical Architecture

Primary Technology Stack:

- **Frontend:** React.js (Component-based UI, State management via Context API/Redux)
- **Backend:** C# ASP.NET Core (3-Tier: Data, Business, Web API)
- **Database:** Microsoft SQL Server (Relational storage, Geography data types)
- **Communication:** SignalR (Real-time updates), RESTful APIs

System Tiers:

- **Presentation Tier (React):** Responsive dashboard for Nurses, Proxies, and Patients.
- **Business Logic Tier (C#):** The "Brain" containing weighted diagnostic algorithms, background workers, and weather logic.
- **Data Access Tier (SQL Server/EF Core):** Entity Framework Core for secure and optimized database transactions.

3. Core Features & Functionality

A. Proxy Caregiver Management

A dedicated dashboard for "Nominated Proxies" (Caregivers) to manage multiple patients. It tracks collection dates for an entire group, allowing for batch collections at clinics.

B. Automated SMS Reminder Service

A C# Background Worker service that executes daily at 08:00 AM. It queries the database for all medication collections due in 48 hours and sends automated SMS notifications to both the Proxy and the Patient.

C. Weighted Symptom Diagnostic Engine

Users input symptoms (Fever, Shaking, etc.). The system calculates a weighted score against known conditions to suggest the most likely issue and provide dosage advice.

D. Geo-Environmental Intelligence

Integration with **OpenWeatherMap API**. The system provides proactive health advice based on local climate (e.g., advising asthma patients during high-wind days in Gqeberha).

E. Smart Clinic Finder

Uses **Google Maps API** to locate the nearest clinic. Features include real-time "Open/Closed" status based on server-time checks against clinic business hours.

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4. Database Design (SQL Schema)

Table Name	Primary Fields	Description
Users	UserID, Username, PasswordHash, Role (Nurse/Proxy/Patient)	System-wide authentication and RBAC.
Patients	PatientID, FullName, IDNumber, ProxyID (FK), ClinicID (FK)	Profiles of people receiving care.
Clinics	ClinicID, Name, Latitude, Longitude, OpenTime, CloseTime	GIS data for the Clinic Finder feature.
Collections	CollectionID, PatientID (FK), ScheduledDate, Status (Pending/Collected)	Tracks chronic medication cycles.
SymptomMap	SymptomID, ConditionID, Weight (1-10)	The core mapping for the diagnostic engine.

5. The Logic "Brain": Weighted Algorithm

Below is a conceptual example of the C# Service that handles the diagnosis:

```
public class DiagnosisResult {
  public string ConditionName { get; set; }
  public int TotalScore { get; set; }
}

// Logic: Calculate total weight for each condition based on selected symptoms
var scores = _context.SymptomMaps
  .Where(sm => userSymptomIds.Contains(sm.SymptomId))
  .GroupBy(sm => sm.Condition.Name)
  .Select(g => new DiagnosisResult {
    ConditionName = g.Key,
    TotalScore = g.Sum(x => x.Weight)
  })
  .OrderByDescending(r => r.TotalScore);
```

6. Future Roadmap: AI Vision Assessor

Phase 2 involves integrating **Azure AI Vision**. This will allow users to upload images of symptoms (e.g., skin rashes) or video of physical tremors. The system will use OCR to read medication labels to ensure the correct dosage is being logged, reducing human error.

7. Security & Compliance (POPIA)

- **Data Encryption:** All patient data is encrypted at rest using SQL TDE.
- **Privacy:** No sensitive medical images are stored on the server; analysis happens in volatile memory.
- **Audit Logs:** Every medication collection log is timestamped with the Nurse ID for accountability.

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